MIL-M-28787/261(EC)
AMENDMENT 1
JUL 6 1983

MILITARY SPECIFICATIONS

MODULES, STANDARD ELECTRONIC

GATE, INVERTER, DIGITAL, KEY CODE EMM

This amendment forms a part of Military Specification MIL-M-28787/261(EC), dated 1 June 1982, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, AND 6

TABLE I: Delete and substitute new table I as printed on pages 2, 3, and 4 of this amendment.

PAGE 7

4.5.1, eighth line: Delete "100°C" and substitute "85°C".

PAGE 8

6.4, second line: Delete "100°C" and substitute "85°C".

6.4, Output drive current (V_{DD} = 5 V): Delete "100°C" (two places) and substitute "85°C" (two places).

PAGE 22

FIGURE 8, NOTE 2: Delete "100°C" and substitute "85°C".

PAGE 25

FIGURE 9, NOTE 2: Delete "100°C" and substitute "85°C".

Agent:
DLA - ES

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Navy - EC

(Project 5963-N501)

FSC 5963

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	S	10	~	10	~	10	~	10	~	10	~	MΩ
	End-of-life	II	S	1	~	1	~	1	~	1	~	1	~	MΩ
B1-B18	Input clamp diode voltage, negative	III	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C18	Input clamp diode voltage, positive	IV	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D18	Output voltage, high level, I_{OH} = 0	V	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
E1-E18	Output voltage, low level, I_{OL} = 0	VI	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F18	Output voltage, high level, V_{DD} = 5 V	VII	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
G1-G18	Output voltage, low level, V_{DD} = 5 V	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H18	Output voltage, high level, V_{DD} = 15 V	IX	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
J1-J18	Output voltage, low level, V_{DD} = 15 V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
K1-K18	Input current, high level	XI	X	0	200	---	---	0	200	---	---	0	200	nA
L1-L18	Input current, low level	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
M1	Power supply current, low level	XIII	X	0	6	0	6	0	60	0	6	0	98	µA
M2	Power supply current, high level	XIII	X	0	6	0	6	0	60	0	6	0	98	µA
N1-N18	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XIV	S	20	200	20	200	22	220	15	150	23	236	ns
N19-N36	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XIV	0	68	680	68	680	75	750	51	515	80	803	ns
P1-P18	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XV	S	20	200	20	200	22	220	15	150	23	236	ns
P19-P36	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XV	0	68	680	68	680	75	750	51	515	80	803	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q18	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XVI	Q	10	100	10	100	11	110	7	75	12	119	ns
R1-R18	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XVII	Q	10	100	10	100	11	110	7	75	12	119	ns
S1-S18	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XVIII	Q	8	80	8	80	9	90	6	60	10	96	ns
T1-T18	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XIX	Q	8	80	8	80	9	90	6	60	10	96	ns
U1-U18	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XX	S	12	125	12	125	14	140	9	95	15	149	ns
U19-U36	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XX	Q	32	325	32	325	36	360	24	245	39	385	ns
V1-V18	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXI	S	12	125	12	125	14	140	9	95	15	149	ns
V19-V36	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXI	Q	32	325	32	325	36	360	24	245	39	385	ns
W1-W18	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXII	Q	8	80	8	80	9	90	6	60	10	96	ns
X1-X18	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXIII	Q	8	80	8	80	9	90	6	60	10	96	ns
Y1-Y18	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXIV	Q	6	60	6	60	6	65	4	45	7	71	ns
Z1-Z18	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXV	Q	6	60	6	60	6	65	4	45	7	71	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA	Power supply filter capacitance	XXVI	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
BB1	C2 capacitance	XXVII	X	176	224	176	224	176	224	176	224	176	224	pF
BB2	C3 capacitance	XXVII	X	176	224	176	224	176	224	176	224	176	224	pF
CC1-CC18	Input capacitance	XXVIII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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